

HDOC 12

Name: Keshav Mittal

SAP ID: 590032934

Batch: 16

1. Algorithm

1. Start
2. Declare variables for choice, num, k, temp, rem, first_digit, last_digit, sum, prod, count_less, count_equal, count_greater.
3. Display the menu options to the user.
4. Read the user's choice.
5. If choice is 1 or 2, prompt the user to input a positive integer num. Validate that $\text{num} > 0$.
6. Use a switch statement based on the value of choice.
7. Case 1 (Sum and Product of first and last digits):
 - Calculate $\text{last_digit} = \text{num} \% 10$.
 - Set $\text{temp} = \text{num}$.
 - Loop while ($\text{temp} \geq 10$): set $\text{temp} = \text{temp} / 10$.
 - After the loop, set $\text{first_digit} = \text{temp}$.
 - Calculate $\text{sum} = \text{first_digit} + \text{last_digit}$ and $\text{prod} = \text{first_digit} * \text{last_digit}$.
 - Display sum and prod, then break.
8. Case 2 (Count digits compared to k):
 - Prompt the user to input a digit k.
 - Initialize $\text{count_less} = 0$, $\text{count_equal} = 0$, $\text{count_greater} = 0$.

- Set temp = num.
- Loop while (temp > 0):
 - Calculate rem = temp % 10.
 - If rem < k, increment count_less.
 - Else if rem == k, increment count_equal.
 - Else, increment count_greater.
 - Update temp = temp / 10.
- Display count_less, count_equal, and count_greater, then break.

9. Default Case: Display an "Invalid choice" error message.

10. Stop

2. Test Case Table

TC ID	Scenario Description	Inputs (choice, num, k)	Expected Output
TC-01	Valid inputs for Menu Option 1	Choice: 1 Num: 12345	Sum = 6 Product = 5
TC-02	Valid inputs for Menu Option 2	Choice: 2 Num: 12345 k: 3	Count_less: 2 Count_equal: 1 Count_greater: 2
TC-03	Single digit number (Option 1)	Choice: 1 Num: 7	Sum = 14 Product = 49
TC-04	All digits same as k (Option 2)	Choice: 2 Num: 5555 k: 5	Count_less: 0 Count_equal: 4 Count_greater: 0
TC-05	Invalid negative number input	Choice: 1 Num: -500	Error: Please enter a strictly positive integer.
TC-06	Invalid menu choice	Choice: 4	Invalid choice. Please choose 1 or 2.

3. C Program Source Code

```
#include<stdio.h> int main() {
    int choice;
    long long num, temp;
    int k, rem, first_digit, last_digit; int sum = 0, prod = 1;
    int count_less = 0, count_equal = 0, count_greater = 0;

    // Display Menu
    printf("--- HDOC 12 Operations Menu ---\n");
    printf("1. Find sum and product of first and last digits\n"); printf("2. Count digits lesser than, equal to, and
    greater than k\n"); printf("Enter your choice (1 or 2): ");
    scanf("%d", &choice);

    // Validate and Take Input
    if (choice == 1 || choice == 2) { printf("Enter a positive integer:
        "); scanf("%lld", &num);

        if (num <= 0) {
            printf("Error: Please enter a strictly positive integer.\n"); return 1; // Exit program due to
            invalid input
        }
    }

    // Switch Case Logic switch (choice) {
        case 1:
            last_digit = num % 10; temp = num;

            // Loop to find the first digit while (temp >= 10) {
                temp /= 10;
            }
            first_digit = temp;

            sum = first_digit + last_digit; prod = first_digit *
            last_digit;

            printf("Number: %lld\n", num);
            printf("Sum of first and last digit: %d\n", sum);
```

```
printf("Product of first and last digit: %d\n", prod); break;
```

case 2:

```
printf("Enter the digit k (0-9): "); scanf("%d", &k);
```

```
temp = num;
```

```
while (temp > 0) { rem = temp
```

```
    % 10; if (rem < k) {
```

```
        count_less++;
```

```
    } else if (rem == k) { count_equal++;
```

```
    } else {
```

```
        count_greater++;
```

```
    }
```

```
    temp /= 10;
```

```
}
```

```
printf("Number: %lld, k: %d\n", num, k); printf("Count_lesser:
```

```
%d\n", count_less); printf("Count_equal: %d\n", count_equal);
```

```
printf("Count_greater: %d\n", count_greater); break;
```

default:

```
printf("Invalid choice. Please choose 1 or 2.\n"); break;
```

```
}
```

```
return 0;
```

```
}
```